

SARVAS

Technical Approach

AI-Based Early Warning and Landslide Risk Monitoring System
in the North Eastern Region of India

Problem Statement: 26001 • Ministry of Development of North Eastern
Region • Disaster Management

Pilot districts: Kohima, Nagaland and Aizawl, Mizoram

SARVAS --- Technical Approach

AI-Based Early Warning and Landslide Risk Monitoring System in the North Eastern Region of India

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1. Technical Objective

The system is a multi-source geospatial, spatiotemporal disaster-intelligence platform that estimates landslide hazard, identifies flash-flood-prone areas, evaluates exposed assets, and generates actionable early-warning intelligence.

The technical objective is not to claim deterministic prediction of every landslide at an exact minute and coordinate. Instead, the system estimates:

Static susceptibility: where terrain and environmental conditions are inherently more prone to failure.

Dynamic hazard: whether current and antecedent conditions are capable of triggering a landslide.

Flash-flood hazard: where intense rainfall can generate rapid runoff, water accumulation and flow-path risk.

Compound hazard: where landslide and flood hazards interact.

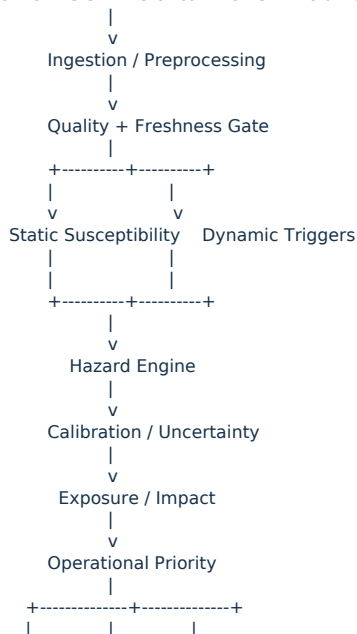
Exposure and impact: which roads, settlements, population and critical infrastructure may be affected.

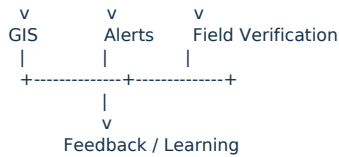
Operational priority: where authorities should inspect, verify, warn or prepare resources first.

Near-real-time alerts: geo-targeted warnings through an authorized verification and dissemination workflow.

The overall architecture is:

Satellite + GIS + Historical Events + Field Reports





2. Fundamental Design: Susceptibility vs Hazard vs Risk

These three concepts are explicitly separated.

2.1 Susceptibility

Answers:

Where could a landslide occur because of the physical characteristics of the location?

Main inputs:

- slope
- elevation
- curvature
- TWI
- TPI
- relief
- geology
- geomorphology
- soil
- land cover
- vegetation
- drainage characteristics
- historical landslide evidence

2.2 Dynamic hazard

Answers:

Are current environmental conditions capable of triggering failure?

Main inputs:

- rainfall
- rainfall intensity
- rainfall duration
- antecedent rainfall
- soil moisture
- flood condition
- Sentinel-1 change/deformation evidence
- verified field observations

2.3 Exposure / impact

Answers:

If a hazardous area fails, what can be affected?

Main inputs:

- population
- settlements
- roads
- bridges
- buildings
- hospitals
- schools
- utilities
- critical infrastructure
- accessibility and alternative routes

Therefore:

SUSCEPTIBILITY + TRIGGER = HAZARD

HAZARD + EXPOSURE + CRITICALITY = OPERATIONAL PRIORITY

Roads and population are not automatically physical landslide causes. They are primarily exposure/impact variables unless engineering-quality road-cut or drainage information is available.

3. Pilot Geography and Common Analysis Grid

3.1 Districts

The current pilot covers:

- Kohima District, Nagaland
- Aizawl District, Mizoram

The architecture is designed to scale to the complete North Eastern Region.

3.2 District boundaries

District boundaries are stored as geospatial polygons and normalized to WGS84 ([EPSG:4326](#)) for interoperability with web GIS.

3.3 Analysis grid

The project uses an approximately 500 m × 500 m analysis grid.

Current grid:

- Kohima: 6,055 cells
- Aizawl: 10,906 cells
- Total: 16,961 cells

The 500 m grid is the common analysis unit, not the native resolution of every sensor.

This distinction is mandatory.

Example:

SRTM ~30 m

Sentinel-2 10/20/60 m depending on band
 GPM ~0.1° (~10 km)
 SMAP ~9 km
 Project ~500 m analysis cell

A coarse satellite observation can therefore be mapped to many project cells, but the system must never claim that the coarse sensor has acquired 500 m native measurements.

4. Data-Source Architecture

Layer	Source Data	Purpose
Rainfall	NASA GPM IMERG precipitation rate/accumulation	Dynamic landslide + flash-flood Early trigger
Soil moisture	NASA SMAP SPL3SMP_E soil moisture Antecedent wetness V006	
Terrain	SRTM DEM	Static terrain
SAR	Copernicus VV/VH/change/coherence where suitable	Surface-change/deformation/flood Sentinel-1 evidence
Vegetation	Copernicus NDVI/NDMI/land-cover features	Environmental/susceptibility Sentinel-2 MSI condition
Geology	GSI / authoritative lithology/structure	Susceptibility layers
Geomorphology	GSI / authoritative geomorphic units	Susceptibility layers
Soil	authoritative soil texture/depth/drainage etc.	Susceptibility + hydrology data
Hydrology	DEM + hydrography flow accumulation/drainage/HAND	Flood + landslide conditioning
Historical events	GSI/state event location/date/type	Labels and susceptibility evidence agencies/verified sources
Roads	OSM / authoritative road network/class/bridges	Exposure and possible GIS anthropogenic conditioning
Population	official/gridded population density	Exposure population data
Settlements	OSM / authoritative settlements/buildings	Exposure GIS
Infrastructure	authoritative hospitals/schools/bridges/utilities	Exposure GIS/OSM where appropriate
Field reports	authorized users observations/photos/location	Verification/corroboation

5. Data Ingestion Layer

Each source follows:

```

Source API / file
|
v
Download / query
|
v
Raw archive
|
v
Format conversion
|
v
Geospatial reprojection
|
v
Quality control
|
v
Feature extraction
|
v
500 m cell mapping
|
v
Feature store
  
```

Every observation stores:

source

product
version
acquisition timestamp
ingestion timestamp
native resolution
temporal resolution
quality flag
data status
processing version

This provides complete data provenance.

6. NASA GPM IMERG Rainfall Pipeline

6.1 Why GPM

Rainfall is one of the most important dynamic triggers for rainfall-induced landslides and flash floods.

A single rainfall value is insufficient. The system requires:

- intensity
- duration
- accumulation
- antecedent rainfall
- short-term and long-term rainfall history

6.2 Product

Primary rainfall source:

NASA GPM IMERG Early Run
Product family: IMERG / 3IMERG
Temporal resolution: 30 minutes
Spatial grid: approximately $0.1^\circ \times 0.1^\circ$
Native spatial scale: approximately 10 km

IMERG Early is designed for low-latency applications. It should be described as near-real-time, not instantaneous.

6.3 Earthdata authentication

The ingestion pipeline uses NASA Earthdata authentication.

Credentials must remain local:

.env
environment variable
secure secret storage

Credentials must never be:

- hard-coded
- committed to Git
- exposed in the frontend
- pasted into public documentation

6.4 Rainfall features

For every analysis cell and prediction timestamp:

rain_30m
rain_1h
rain_3h
rain_6h
rain_12h
rain_24h
rain_48h
rain_72h
rain_7d

```

peak_intensity
rainfall_duration
antecedent_rainfall

```

6.5 Rolling accumulation

For half-hour rainfall observations:

$$R_T = \sum \text{precipitation}_i$$

where all observations inside time window **T** are summed.

Examples:

```

R_3h
R_6h
R_24h
R_72h
R_7d

```

6.6 Rainfall intensity

The system preserves both:

rainfall rate

and:

rainfall accumulation

because:

100 mm in 2 hours

is not hydrologically equivalent to:

100 mm in several days

6.7 Rainfall duration

Duration is derived from the sequence of rainfall observations and a configurable intensity threshold.

Conceptually:

```

duration =
continuous time with rainfall >= configured threshold

```

The threshold must eventually be learned/calibrated from local historical events.

6.8 Antecedent rainfall

Antecedent rainfall represents prior wetting:

```

AR_3d
AR_7d

```

and potentially longer windows.

This is important because a heavy burst following already-wet conditions can have a different effect from the same burst after a dry period.

6.9 Current pipeline status

The prototype has successfully ingested real NASA GPM IMERG Early observations.

The current pipeline has:

- discovered recent NASA granules;
- selected newest granules rather than oldest records;
- mapped GPM observations to all project cells;
- applied freshness validation;
- preserved native-resolution metadata.

Current implementation maps approximately 67 distinct GPM macro-pixels to the 16,961 project cells.

Therefore:

16,961 cells does not mean 16,961 independent GPM rainfall measurements.

7. NASA SMAP Soil-Moisture Pipeline

7.1 Why soil moisture

Rainfall represents forcing.

Soil moisture represents the antecedent hydrological state.

The combination:

high recent rainfall
+
already wet soil

can indicate increased trigger conditions.

7.2 Product

Primary source:

NASA SMAP
SPL3SMP_E
Version 006
Enhanced L3 Radiometer Global Daily
EASE-Grid 2.0
approximately 9 km
daily

Primary AM soil-moisture variable:

/Soil_Moisture_Retrieval_Data_AM/soil_moisture

Units:

m³/m³

The PM product/fields are handled where available and appropriate.

7.3 Features

soil_moisture_current
soil_moisture_previous
soil_moisture_change
soil_moisture_change_percent
quality_flag
observation_timestamp
data_age

7.4 Change

$\Delta_{SM} = SM_{current} - SM_{previous}$

Percentage change:

$\Delta_{SM_percent} = 100 \times (SM_{current} - SM_{previous}) / \text{abs}(SM_{previous})$

with protection against invalid or zero denominators.

7.5 Spatial mapping

SMAP is approximately 9 km native resolution.

The project maps those observations to the 500 m analysis cells.

The system must display:

Native SMAP resolution: ~9 km
Analysis grid: ~500 m

rather than implying 500 m SMAP resolution.

7.6 Freshness

Every SMAP observation receives:

AVAILABLE
STALE
UNAVAILABLE
REQUIRES_AUTH
INVALID

A stale observation is not silently treated as current.

The current pilot has real NASA SMAP observations populated across the project grid, but the latest cached observation may be stale depending on ingestion time. The freshness gate therefore determines whether SMAP is allowed to contribute to a current dynamic assessment.

8. SRTM DEM and Terrain Pipeline

8.1 DEM

Primary terrain source:

USGS/NASA SRTM
approximately 30 m

8.2 Terrain derivatives

The DEM is processed into:

elevation
slope
aspect
curvature
TPI
TRI
relief
TWI
flow direction
flow accumulation
drainage density
distance to drainage
HAND

8.3 Slope

Slope is calculated from local elevation gradients.

Conceptually:

slope =
 $\text{atan}(\sqrt{(dz/dx)^2 + (dz/dy)^2})$

The system stores slope in degrees and can additionally derive percent slope.

8.4 Aspect

Aspect represents direction of slope orientation.

It may influence environmental characteristics such as:

- solar exposure;
- vegetation;
- evapotranspiration;
- moisture persistence.

Aspect should not receive an arbitrary universal danger score.

8.5 Curvature

Curvature identifies:

concave terrain
convex terrain
approximately planar terrain

It is useful for representing local water convergence/divergence and morphology.

8.6 TWI

Topographic Wetness Index:

$$TWI = \ln(a / \tan(\beta))$$

where:

- **a** = specific catchment area;
- **beta** = local slope angle.

TWI is used as a terrain-derived wetness/convergence indicator.

8.7 TPI

Topographic Position Index compares local elevation with surrounding terrain.

It can help distinguish:

ridge
upper slope
mid-slope
valley
depression

8.8 Flow accumulation

Flow accumulation identifies where DEM-derived runoff converges.

It is particularly important for the flash-flood module.

9. Geological Susceptibility Layer

9.1 Why geology

Slope stability depends on:

- lithology;
- rock strength;
- weathering;
- fractures;
- bedding;
- joints;
- faults;
- lineaments;
- permeability;
- regolith/soil thickness.

9.2 Features

Where authoritative datasets are available:

lithology
geological unit

weathering
 fault distance
 lineament density
 structural orientation
 geological contacts

are spatially joined to the 500 m cells.

9.3 Source priority

Preferred hierarchy:

GSI / government authoritative source
 >
 official state GIS
 >
 validated scientific dataset
 >
 open supplementary source

No geological value is fabricated when coverage is unavailable.

10. Geomorphology

Geomorphological units describe terrain-forming processes.

Possible source categories include:

structural hills
 dissected hills
 valleys
 terraces
 alluvial surfaces
 debris/scree
 erosional landforms

The actual categories must be derived from the source dataset.

Features:

geomorphology_class
 geomorphic_distance
 landform_type

are encoded for susceptibility.

11. Soil Properties

Where reliable spatial soil data are available:

soil_texture
 sand_fraction
 silt_fraction
 clay_fraction
 soil_depth
 bulk_density
 hydraulic_conductivity
 permeability
 drainage_class
 available_water_capacity
 organic_matter

can be added.

Why:

rainfall
 |
 +--> infiltration
 |
 +--> soil saturation
 |
 +--> runoff

Soil properties therefore influence both landslide conditioning and flood runoff.

12. Sentinel-2 Vegetation Pipeline

12.1 Why vegetation

Vegetation can influence:

- root reinforcement;
- interception;
- evapotranspiration;
- surface erosion;
- soil exposure;
- land-use change.

Vegetation is therefore a conditioning/environmental feature.

It is not valid to use a simplistic rule:

low NDVI = landslide

without local evidence.

12.2 Satellite

Primary optical source:

Copernicus Sentinel-2 MSI

Band resolution:

10 m / 20 m / 60 m
depending on band

12.3 NDVI

For Sentinel-2:

$$\text{NDVI} = (\text{B8} - \text{B4}) / (\text{B8} + \text{B4})$$

where:

B8 = near infrared

B4 = red

12.4 Additional vegetation features

Where data quality permits:

NDVI

NDMI

EVI

vegetation fraction

seasonal vegetation anomaly

vegetation change

12.5 Cloud handling

Before extracting optical features:

cloud mask

cirrus mask

invalid-pixel mask

are applied.

Cloud-contaminated observations are not treated as valid vegetation measurements.

12.6 Vegetation change

For two valid observations:

Delta_NDVI =

$NDVI_{current} - NDVI_{reference}$

A persistent decline can be used as environmental-change evidence, but must be corroborated before being interpreted as landslide evidence.

13. Sentinel-1 SAR Pipeline

13.1 Why Sentinel-1

Sentinel-1 uses C-band synthetic aperture radar.

SAR is valuable because it can operate:

- day and night;
- through cloud cover;
- during monsoon conditions.

13.2 Features

Possible features:

VV backscatter
 VH backscatter
 VV/VH ratio
 VV temporal change
 VH temporal change
 coherence
 interferometric deformation
 deformation velocity
 deformation acceleration
 change confidence

13.3 Backscatter change

For two observations:

$\Delta_{VV} = VV_{current} - VV_{previous}$

and:

$\Delta_{VH} = VH_{current} - VH_{previous}$

13.4 InSAR

For suitable acquisitions:

```

SLC t1
+
SLC t2
|
v
Interferogram
|
v
Phase processing
|
v
Unwrapping
|
v
Atmospheric/orbit correction
|
v
LOS deformation
  
```

This can provide evidence of pre-failure ground movement.

13.5 Corroboration requirement

SAR change is not automatically a landslide.

Backscatter changes may result from:

- vegetation;
- soil moisture;
- surface roughness;
- construction;
- agriculture;
- flooding;
- geometric effects.

Therefore SAR is treated as corroborating evidence.

13.6 Current pilot status

The project has a Sentinel-1 acquisition/indexing architecture, but actual scene downloads/processing require the corresponding external data access/authentication.

The system therefore uses:

PENDING
REQUIRES_AUTH
UNAVAILABLE

rather than fabricating SAR values.

14. Historical Landslide Inventory

14.1 Sources

Historical events should be compiled from:

- Geological Survey of India;
- state disaster-management agencies;
- government reports;
- verified field reports;
- validated remote-sensing interpretation.

14.2 Event schema

event_id
date
time
latitude
longitude
district
state
source
source_reference
landslide_type
trigger
confidence
geometry
description

14.3 Label quality

Each record receives:

VERIFIED
NEEDS_VERIFICATION
APPROXIMATE
MISSING_COORDINATE

Only sufficiently reliable events become hard positive labels.

14.4 No false negatives

This rule is critical:

no recorded landslide

does not mean:

confirmed no landslide

An unobserved location is unknown.

Therefore the system must not randomly label all other cells as negatives.

15. Feature Engineering

The final feature dataset is organized around:

cell_id × prediction_timestamp

Static features

elevation
slope
aspect
curvature
TPI
TRI
TWI
relief
lithology
geomorphology
soil properties
land cover
NDVI
vegetation change
distance to drainage
drainage density
lineament/fault features
historical event density

Dynamic features

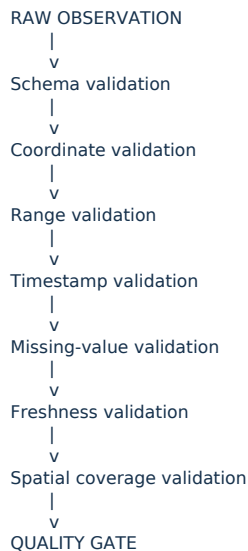
rain_30m
rain_1h
rain_3h
rain_6h
rain_12h
rain_24h
rain_48h
rain_72h
rain_7d
peak rainfall intensity
rainfall duration
antecedent rainfall
soil moisture current
soil moisture previous
soil moisture change
soil moisture change %
flood indicator
Sentinel-1 change
deformation/coherence where valid
field verification

Exposure features

road distance
road length in hazard area
road class
bridge exposure
population density
settlement density
building count
hospital distance
school distance
critical infrastructure count

16. Data Quality and Freshness Gate

Before any model inference:



16.1 Data statuses

AVAILABLE
 STALE
 UNAVAILABLE
 REQUIRES_AUTH
 INVALID
 PARTIAL

16.2 Missing values

Never perform:

missing rainfall -> 0
 missing soil moisture -> 0
 missing SAR -> no change

Instead:

value = NULL
 status = UNAVAILABLE

This prevents the model from confusing missing data with safe conditions.

17. Static Susceptibility Model

The first model stage answers:

Where is the terrain inherently susceptible?

A transparent baseline can be expressed as:

TSI =
 w1*slope
 +w2*TWI
 +w3*curvature
 +w4*relief
 +w5*geology
 +w6*geomorphology
 +w7*soil
 +w8*landcover
 +w9*historical_evidence

All features are normalized before combination.

Weights must be one of:

- learned from adequate training data;
- justified by an authoritative methodology;
- explicitly documented prototype policy weights.

They must not be described as ML-learned if they were manually selected.

18. Positive-Unlabeled Learning

The pilot historical dataset has very few verified positive cells and does not contain a reliable set of confirmed negatives.

Therefore:

Positive = verified landslide
Unknown = not confirmed

not:

Unknown = negative

A Positive-Unlabeled or anomaly/similarity approach can:

- learn environmental similarity to known event cells;
- identify cells with similar susceptibility conditions;
- combine that evidence with terrain susceptibility;
- avoid falsely claiming that unobserved cells are safe.

This is appropriate as an interim architecture while the verified inventory is expanded.

19. Dynamic Trigger Engine

The dynamic engine converts current conditions into trigger evidence.

Prototype trigger inputs:

Rainfall
Soil moisture
Flood
Sentinel-1
Field verification

A transparent prototype configuration can be:

Rainfall	45%
Soil moisture	35%
Flood	10%
Sentinel-1 SAR	5%
Field verification	5%

Therefore:

$$\begin{aligned} \text{DynamicScore} = & \\ & 0.45 * \text{RainScore} \\ & + 0.35 * \text{SoilScore} \\ & + 0.10 * \text{FloodScore} \\ & + 0.05 * \text{SARScore} \\ & + 0.05 * \text{VerificationScore} \end{aligned}$$

These are prototype policy weights, not calibrated probabilities.

Only quality-approved, available observations contribute.

20. Rainfall Trigger Normalization

Rainfall should not use one arbitrary universal threshold.

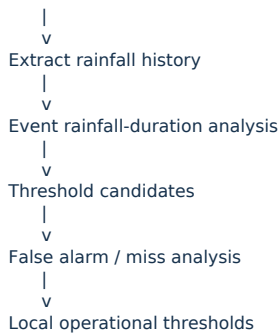
Instead, derive several normalized indicators:

I_30m
I_3h
I_6h
I_24h
I_72h
I_7d

and eventually estimate local rainfall-duration relationships.

Conceptual workflow:

Historical events



Possible future methods:

Intensity-Duration threshold
 Cumulative rainfall threshold
 Antecedent rainfall threshold
 Intensity-Duration-Antecedent model

Thresholds must be locally validated for NER.

21. Soil-Moisture Trigger

Soil moisture can be converted into normalized evidence using:

current percentile
 historical percentile
 change %
 antecedent wetness

Conceptually:

SMScore =
 f(
 current moisture,
 moisture anomaly,
 rate of change,
 rainfall context
)

A high soil-moisture value alone does not prove landslide occurrence.

It is a trigger/conditioning signal.

22. Dynamic Hazard Combination

The prototype combines:

Static Susceptibility
 +
 Dynamic Trigger Evidence

Example:

CombinedHazard =
 0.50*StaticSusceptibility
 +
 0.50*DynamicTrigger

This is a transparent prototype policy.

Once enough validated training data exist, the manually defined combination can be replaced by a learned model.

23. Machine-Learning Hazard Model

23.1 Candidate algorithms

For tabular geospatial features:

XGBoost
 LightGBM
 Random Forest

HistGradientBoosting

The first production candidate should normally be a strong interpretable tree-based model rather than an unnecessarily complex deep-learning system.

23.2 Input

```
X_t =
{
terrain,
geology,
geomorphology,
soil,
vegetation,
rainfall history,
soil moisture,
SAR,
flood,
season,
land cover,
historical evidence
}
```

23.3 Target

Potential target:

```
Y_t =
1 if a validated landslide event occurs
0 if a validated non-event/background observation is confirmed
```

Until reliable negative/background labels exist, use PU learning or another appropriate weakly supervised strategy.

24. Probability Calibration

A model score is not automatically a probability.

Do not say:

```
model = 0.80
therefore 80% probability
```

unless calibration has been demonstrated.

Candidate calibration methods:

```
Platt scaling
Isotonic regression
Beta calibration
```

Calibration must use independent validation data.

25. Spatial Validation

Random splitting can produce overly optimistic results because nearby grid cells are spatially correlated.

Use:

Spatial block cross-validation

For example:

5 km × 5 km geographic blocks

Train on some blocks and test on geographically separated blocks.

26. Temporal Validation

The model must not use future information to predict the past.

Example:

Training:

older event years

Testing:
later event year

The exact split depends on the final validated inventory.

All features must satisfy:

`feature_timestamp <= prediction_timestamp`

27. Cross-District Generalization

Test geographic transferability:

Train -> Kohima
Test -> Aizawl

and:

Train -> Aizawl
Test -> Kohima

This measures whether the model is learning generalizable relationships rather than memorizing one district.

28. Evaluation Metrics

Use:

Precision
Recall
F1
ROC-AUC
PR-AUC
Brier score
Calibration error
False Alarm Rate
Miss Rate
Lead Time
Spatial Hit Rate

For rare-event landslides, PR-AUC and event-level recall can be especially informative.

Do not report a generic accuracy number without a scientifically valid validation design.

29. Landslide Probability Output

After adequate training and calibration:

$P(\text{Landslide} \mid X_t)$

can be produced.

The probability should be interpreted only within the validated population and forecast horizon.

Before calibration:

probability = NULL
calibration_status = NOT_CALIBRATED

This is preferable to inventing a calibrated percentage.

30. Flash-Flood Module

The platform contains a separate flood-hazard pipeline because landslides and flash floods can share rainfall forcing but have different physical mechanisms.

The flash-flood module estimates:

flash-flood susceptibility
+
rainfall-triggered runoff hazard
+
flow-path concentration

+
possible inundation

It should not claim exact street-level flood depth unless hydraulic observations/model calibration support that output.

31. Flash-Flood Static Susceptibility

Primary features:

elevation
slope
flow accumulation
flow direction
TWI
drainage density
distance to stream
catchment area
valley geometry
HAND
soil infiltration properties
land cover
imperviousness where available
stream network

The most important conceptual process is:

```

Rainfall
|
v
Infiltration / runoff
|
v
Flow accumulation
|
v
Drainage network
|
v
Downstream concentration
  
```

32. DEM-Based Hydrological Processing

The flood module uses:

```

DEM
|
v
Hydrologic conditioning
|
v
Flow direction
|
v
Flow accumulation
|
v
Stream extraction
|
v
Catchment delineation
|
v
Drainage density
|
v
HAND
  
```

These layers are then mapped to the common analysis grid.

33. HAND --- Height Above Nearest Drainage

Conceptually:

HAND =
elevation(cell)
-
elevation(reference drainage)

Low HAND generally identifies terrain closer vertically to drainage channels.

It is useful for flood susceptibility but is not, by itself, a complete flood-depth model.

34. Flash-Flood Rainfall Trigger

The flood trigger uses:

```
rain_30m
rain_1h
rain_3h
rain_6h
rain_12h
rain_24h
rain_72h
peak intensity
rainfall duration
antecedent rainfall
```

Conceptually:

```
FlashFloodTrigger =
f(
short-duration intensity,
rainfall accumulation,
antecedent wetness,
catchment response,
flow concentration
)
```

This is more physically meaningful than using rainfall alone.

35. Rainfall-Runoff Model

For an advanced flood system:

```
Rainfall
|
v
Interception
|
v
Infiltration
|
v
Surface runoff
|
v
Channel routing
|
v
Water accumulation
|
v
Flood extent/depth
```

Candidate hydrological approaches:

```
SCS Curve Number
Green-Ampt infiltration
Unit hydrograph
Kinematic-wave runoff
Distributed hydrological model
```

The final choice depends on the availability of:

- soil;
- land cover;
- rainfall;
- drainage;
- channel geometry;
- gauge/stream observations;
- historical flood extents.

36. Flash-Flood ML Model

Potential inputs:

rainfall intensity
rainfall accumulation
antecedent rainfall
soil moisture
flow accumulation
HAND
TWI
slope
drainage density
distance to stream
catchment area
land cover
imperviousness
soil
historical flood labels

Targets can be:

flood / no flood

or, if enough observations exist:

inundation depth
inundation extent
arrival time

37. Sentinel-1 Flood Detection

SAR can help detect flood extent even under cloud cover.

Workflow:

```

Pre-event SAR
+
Post-event SAR
|
v
Radiometric calibration
|
v
Speckle filtering
|
v
Terrain correction
|
v
Backscatter comparison
|
v
Water classification
|
v
Flood polygon
|
v
500 m grid intersection
|
v
flood_indicator
  
```

The flood layer can then feed:

FlashFloodHazard

and, where relevant:

LandslideDynamicContext

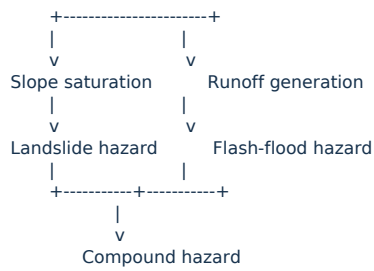
38. Compound Landslide + Flood Hazard

The system explicitly supports compound events.

Example:

```

Extreme rainfall
|
  
```



A mountainous road corridor can therefore have:

landslide-prone slope
 +
 downstream flash-flood corridor
 +
 bridge exposure

which creates a higher operational response requirement.

39. Road Intelligence

Roads have two technical roles.

39.1 Physical conditioning

If detailed road engineering data exist, road cuts can affect:

- slope geometry;
- drainage;
- toe support;
- runoff concentration.

Such engineering features can be included in susceptibility.

39.2 Exposure

For operational impact:

road_distance
 road_length_inside_hazard_zone
 road_class
 bridge_count
 alternative_route_availability

are exposure variables.

Example:

same hazard score
 +
 major highway

has different operational consequences from:

same hazard score
 +
 unoccupied remote slope

40. OpenStreetMap Road Pipeline

Where appropriate, OSM can provide:

motorway
 trunk
 primary
 secondary
 tertiary
 residential
 track
 bridge
 tunnel

The system can calculate:

```
nearest_road_distance
road_length_intersecting_hazard
road_class
bridge_exposure
```

OSM coverage may be incomplete, so the dataset must retain source and completeness metadata.

41. Population and Settlement Exposure

Population is not a landslide cause.

It is an exposure variable.

Features:

```
population_density
population_inside_hazard_area
settlement_density
building_count
distance_to_settlement
```

Potential output:

Estimated population exposed

only when reliable population data are available.

42. Critical Infrastructure

Potential assets:

```
Hospitals
Schools
Bridges
Police stations
Fire stations
Government offices
Power infrastructure
Water infrastructure
Telecommunication infrastructure
Relief shelters
Major roads
Railways
Airports
```

Each asset can store:

```
asset_id
asset_type
latitude
longitude
criticality
nearest_hazard_cell
hazard_score
exposure_status
```

43. Exposure / Impact Model

The impact layer is separate from the physical hazard model.

Conceptually:

```
Impact =
f(
  hazard,
  population,
  road importance,
  infrastructure criticality,
  accessibility,
  isolation,
  alternative routes
)
```

A simple conceptual relationship is:

OperationalPriority

=
Hazard × Exposure × Criticality

The exact implementation must be documented and validated.

44. Area-Level Risk Intelligence

Users should be able to search:

Kohima
Aizawl
Durtlang
Khonoma
Bara Bazar
Treasury

The search system must be cross-district.

A search for an Aizawl locality should:

```

graph TD
    A[identify locality] --> B[identify district]
    B --> C[switch map to district]
    C --> D[highlight associated cells]
    D --> E[display risk dashboard]
  
```

The result can show:

hazard
rainfall
soil moisture
terrain
vegetation
flood
SAR
roads
population
infrastructure
historical events
data freshness
recommended action

45. GIS Map Architecture

Frontend:

React
+
MapLibre GL JS

Map layers:

Satellite basemap
District boundary
500 m analysis grid
Susceptibility
Risk
Historical landslides
Rainfall
Soil moisture
Flood
Sentinel-1
Vegetation
Roads
Settlements
Infrastructure
Drainage

Controls:

Zoom
Pan
Search
District switch

Layer toggle
 Cell hover
 Cell click
 Locality selection
 Historical event inspection

46. Cell-Level Explainability

Every selected cell should provide:

Cell ID
 District
 Coordinates

Static susceptibility
 Dynamic trigger
 Combined hazard

Rainfall:
 30 min
 3 h
 24 h
 72 h
 7 d
 intensity
 duration

Soil moisture:
 current
 previous
 change
 freshness

Terrain:
 elevation
 slope
 curvature
 TWI
 TPI
 relief

Vegetation:
 NDVI
 change

SAR:
 VV
 VH
 change
 confidence

Flood:
 indicator
 flow accumulation
 HAND

Exposure:
 road
 population
 infrastructure

Data quality:
 source
 age
 completeness

47. Risk Classification

Prototype risk classes:

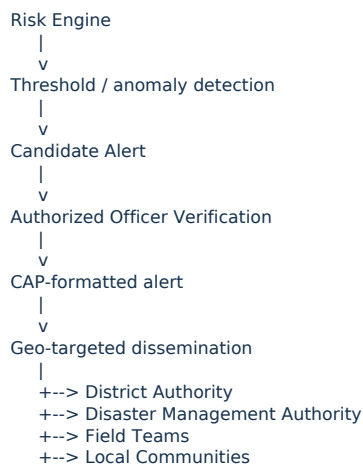
Risk class	Score	Color	Meaning
Critical	≥ 80	Purple	Immediate investigation / urgent advisory
High	60--79.9	Red	Severe hazard attention
Warning	45--59.9	Orange	Early warning / inspection Watch
Enhanced monitoring	30--44.9	Yellow	Enhanced monitoring
Low	< 30	Green	Routine monitoring

These are prototype operational thresholds until calibrated against validated local events.

They must not be presented as universal scientific thresholds.

48. Alert Decision Engine

The alert workflow is:



The prototype should not automatically issue a public evacuation order from an unverified AI score.

49. Near-Real-Time Alert Architecture

The system is designed for:

near-real-time

because satellite products have different acquisition and processing latency.

Examples:

GPM:

30-minute observation cadence, low-latency product

SMAP:

daily observation product

Sentinel-1:

acquisition/revisit dependent

Sentinel-2:

acquisition + cloud-free observation dependent

Field reports:

event-driven

Therefore the dashboard must always display:

last observation

data age

source

freshness status

50. Common Alerting Protocol

The alert payload can use CAP-style fields:

identifier

sender

sent

status

msgType

scope

event

urgency

severity

certainty

effective

expires

area

polygon/circle

instruction

Cell Broadcast integration should be represented as:

integration-ready

until authorized government/telecom infrastructure is actually connected.

The system must not claim direct control over telecom towers.

51. Field Verification

Human verification is a first-class data source.

Possible observations:

fresh ground crack
road settlement
retaining-wall failure
rockfall
mud movement
water seepage
blocked drainage
fresh debris
flooded road
bridge overtopping

Each report stores:

location
timestamp
report type
evidence
reporter role
confidence
verification status

Workflow:

```

graph TD
    A[Field/Citizen Report] --> B[Geolocation]
    B --> C[Evidence validation]
    C --> D[Authorized verification]
    D --> E[Risk evidence update]
    E --> F[Alert decision]
  
```

52. Human + AI Corroboration

The system follows:

```

graph TD
    A[AI detects anomaly] --> B[AI identifies high-risk cell]
    B --> C[Field evidence requested]
    C --> D[Officer verifies]
    D --> E[Alert confidence increases]
  
```

This reduces dependence on any single sensor.

The architecture is therefore:

Predict
+
Verify
+
Corroborate
+

Prioritize
+
Act
+
Learn

53. Uncertainty Model

A mature system should return both:

hazard

and:

uncertainty

Example:

hazard_score = 72
uncertainty = MEDIUM
data_completeness = 68%

Uncertainty sources:

coarse rainfall resolution
coarse soil moisture
missing SAR
cloud contamination
limited historical labels
incomplete road data
incomplete infrastructure data
model extrapolation

A high hazard supported by several independent sensors should be distinguished from a high score generated from sparse evidence.

54. Sensor-Fusion Confidence

For every cell:

Rainfall	AVAILABLE
Soil moisture	AVAILABLE
Terrain	AVAILABLE
SAR	AVAILABLE
Flood	AVAILABLE
Field report	VERIFIED

The system derives an evidence-confidence indicator from:

number of independent sources
freshness
quality
spatial alignment
agreement/disagreement

The UI can show:

Hazard: HIGH
Confidence: HIGH

OR:

Hazard: HIGH
Confidence: LOW

without pretending that confidence itself is a probability.

55. Advanced Spatiotemporal Model

After enough training data are available, the system can evolve from feature snapshots to sequences.

Example:

t-168h
t-144h
t-120h
...
t-24h

t

Each timestamp contains:

rainfall
soil moisture
SAR
flood

while static features remain:

terrain
geology
soil
land cover

A temporal model can then learn:

dry
-> wetting
-> persistent rainfall
-> saturation
-> extreme trigger
-> failure

Candidate models:

LSTM
GRU
Temporal CNN
Temporal Transformer
ConvLSTM
Spatiotemporal GNN

Deep learning should only be introduced when the event inventory supports it.

56. Graph-Based Spatial Modeling

The 500 m grid can be converted to a graph.

```
Cell A ---- Cell B ---- Cell C
|           |           |
Cell D ---- Cell E ---- Cell F
```

Edges can represent:

spatial adjacency
flow direction
drainage connectivity
road connectivity
river connectivity

This is especially useful for flood propagation and upstream/downstream relationships.

57. Leakage Prevention

No future information may enter a historical prediction.

Incorrect:

Predict event at T
using imagery captured after T

Correct:

Prediction at T
uses only information available $\leq T$

Examples:

rainfall(t-72h ... t)
soil moisture(last observation $\leq t$)
SAR(last valid acquisition $\leq t$)
Sentinel-2(last valid cloud-free observation $\leq t$)
terrain(static)

58. Event-Centric Training

For every historical event:

Event location
 +
 surrounding cells
 +
 pre-event temporal window

should be constructed.

Example:

T-168h
 T-120h
 T-72h
 T-48h
 T-24h
 T

For every time step:

rainfall
 soil moisture
 SAR
 flood

are joined with static terrain/geology/vegetation features.

This preserves temporal causality.

59. Training Data Expansion

The pilot cannot remain dependent on a very small number of verified events for a production-grade calibrated probability model.

Training expansion should include:

historical landslide points
 historical landslide polygons
 event date/time
 rainfall history
 soil moisture
 terrain
 geology
 vegetation
 SAR evidence
 flood condition
 land cover
 road context
 field validation

The system should continuously convert verified events into high-quality training examples.

60. Negative / Background Sampling

When sufficient event inventory exists, scientifically defensible background/non-event sampling can be created.

Possible approaches:

verified non-event observations
 carefully sampled background locations
 temporal non-event windows

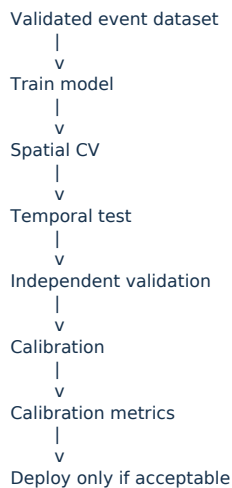
The sampling procedure must be documented.

Do not create negatives by simply assigning:

all non-event cells = 0

without considering observation bias.

61. Model Calibration Pipeline



Calibration metrics:

Brier score
 Expected Calibration Error
 Reliability curve
 Calibration intercept
 Calibration slope

62. Model Explainability

For tree models:

SHAP
 feature importance
 partial dependence

Example explanation:

High recent rainfall
 +
 high antecedent wetness
 +
 steep slope
 +
 high TWI
 +
 supporting SAR change
 =
 elevated hazard evidence

The interface should expose the contributing evidence rather than only a black-box score.

63. Model Output Contract

Example:

```

{
  "cell_id": "KOH_01378",
  "timestamp": "2026-09-18T00:00:00Z",
  "static_susceptibility": 61.48,
  "dynamic_trigger_score": 72.30,
  "hazard_score": 66.89,
  "risk_level": "HIGH",
  "probability": null,
  "calibration_status": "NOT_CALIBRATED",
  "data_quality": "PARTIAL",
  "factors": {
    "rainfall": "HIGH",
    "soil_moisture": "ELEVATED",
    "terrain": "HIGH_SUSCEPTIBILITY",
    "sar": "UNAVAILABLE",
    "flood": "AVAILABLE"
  }
}
  
```

When calibrated:

probability

can be populated.

Before calibration:

probability = null

64. API Architecture

Backend:

Python
FastAPI
Pydantic
GeoPandas
Rasterio
NumPy
Pandas
scikit-learn
XGBoost / LightGBM
PostgreSQL/PostGIS

Core APIs:

GET /api/districts
GET /api/districts/{district}/grid
GET /api/districts/{district}/risk

GET /api/cells/{cell_id}
GET /api/cells/{cell_id}/risk
GET /api/cells/{cell_id}/parameters

GET /api/rainfall/latest
GET /api/soil-moisture/latest
GET /api/satellite/latest
GET /api/flood/latest

GET /api/areas
GET /api/areas/{area_id}/risk

GET /api/model/metadata
GET /api/model/metrics

GET /api/alerts
POST /api/alerts/generate
POST /api/alerts/{id}/verify
POST /api/alerts/{id}/acknowledge
POST /api/alerts/{id}/expire

65. Production Database

Recommended database:

PostgreSQL + PostGIS

Core entities:

districts
analysis_cells
rainfall_observations
soil_moisture_observations
sar_observations
vegetation_observations
flood_observations
landslide_events
roads
settlements
infrastructure
risk_assessments
alerts
field_reports
model_versions

Large raw satellite files should remain in object storage rather than being unnecessarily placed inside relational tables.

66. Data Lake / File Architecture

Recommended:

```
data/
  raw/
  intermediate/
  processed/
  features/
  models/
  validation/
  research/
```

Example:

```
data/raw/gpm/
data/raw/smap/
data/raw/sentinel1/
data/raw/sentinel2/

data/processed/terrain/
data/processed/geology/
data/processed/vegetation/
data/processed/flood/

data/features/
data/models/
data/validation/
```

67. Technology Stack

Frontend

React
Vite
MapLibre GL JS
GeoJSON / vector tiles

Backend

Python
FastAPI
Pydantic
Pandas
NumPy
GeoPandas
Rasterio
Shapely
PyProj
SciPy
scikit-learn
XGBoost / LightGBM

Satellite / geospatial

NASA Earthdata
Earthaccess
HDF5
GDAL
Rasterio
Sentinel processing tools

Database

PostgreSQL
PostGIS

Validation

pytest
schema validation
geospatial validation
data-quality validation
model validation
API tests

68. Security

All external credentials are local-only.

Use:

.env
environment variables
secure secret manager

Protect:

NASA Earthdata token
passwords
API keys
service credentials

Never expose secrets in:

frontend JavaScript
Git
GeoJSON
logs
screenshots
public folders
documentation

69. Pipeline Scheduling

A production deployment can schedule ingestion according to source cadence.

Conceptually:

GPM
-> frequent ingestion

SMAP
-> daily ingestion

Sentinel-1
-> acquisition-driven ingestion

Sentinel-2
-> acquisition/cloud-quality driven ingestion

Field reports
-> event-driven

Static GIS
-> version-driven update

Each ingestion run updates:

last_success_timestamp
last_observation_timestamp
data_age
freshness_status

70. Monitoring

Operational monitoring should track:

GPM last observation
SMAP last observation
Sentinel-1 last acquisition
Sentinel-2 last valid scene
pipeline success/failure
download failures
missing cells
stale observations
API latency
model inference latency
alert count
alert verification time

71. Model Drift

Monitor:

rainfall distribution drift
soil-moisture distribution drift
land-cover changes
vegetation changes
road expansion
urbanization
new infrastructure
sensor/product changes

If drift is detected:

Review
-> Retrain
-> Recalibrate
-> Revalidate
-> Redeploy

72. Reproducibility

Every model release stores:

code version
data versions
feature schema
training period
validation design
hyperparameters
model artifact
calibration artifact
metrics
thresholds
feature importance
software versions

A prediction must be traceable to the input observations that generated it.

73. Fail-Safe Architecture

If a dynamic source is unavailable:

DO NOT:
fabricate data
fill missing rainfall with zero
fill missing SMAP with zero
treat missing SAR as no change
pretend stale data are live
claim calibrated probability

Instead:

dynamic_status = UNAVAILABLE
dynamic_score = NULL
probability = NULL
static_susceptibility = available

The GIS can still show static susceptibility.

74. Operational Decision Layer

AI should provide decision support, not replace authorized disaster-management decisions.

Workflow:

```

AI hazard
|
v
Exposure
|
v
Recommended action
|
v
Authorized authority
  
```

|
v
Warning / response

Possible system recommendations:

MONITOR
INSPECT
VERIFY
RESTRICT ACCESS
ISSUE WARNING
PRE-POSITION RESPONSE TEAMS
PREPARE EVACUATION
EVACUATE

Actual public instructions remain under authorized disaster-management protocols.

75. Priority Queue

If hundreds of cells become hazardous:

Hazard
+
Exposure
+
Criticality
+
Evidence confidence

can be used to create an operational response queue.

Example data:

Priority
Cell
District
Hazard
Population exposed
Road exposure
Critical infrastructure
Confidence
Recommended action

This converts a raw map into an actionable operational dashboard.

76. Complete End-to-End Landslide Pipeline

NASA GPM

- |
- +++> Rainfall rate
- +++> Accumulation
- +++> Intensity
- +++> Duration
- +++> Antecedent rainfall

NASA SMAP

- |
- +++> Current soil moisture
- +++> Previous soil moisture
- +++> Change
- +++> Change %

SRTM

- |
- +++> Elevation
- +++> Slope
- +++> Aspect
- +++> Curvature
- +++> TWI
- +++> TPI
- +++> Relief
- +++> Flow accumulation

GSI / authoritative GIS

- |
- +++> Geology
- +++> Geomorphology
- +++> Structure
- +++> Soil

Sentinel-2

- |
- +++> NDVI
- +++> NDMI
- +++> Vegetation change
- +++> Land-cover information

Sentinel-1

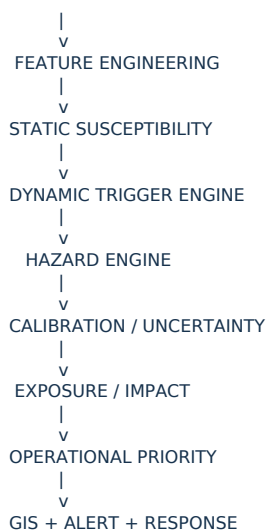
- |
- +++> VV
- +++> VH
- +++> Backscatter change
- +++> Coherence/deformation where valid

Historical inventory

- |
- +++> Verified events
- +++> Event timing
- +++> Event location

Field reports

- |
- +++> Human verification
- +++> Evidence
- +++> Confirmation



77. Complete Flash-Flood Pipeline

NASA GPM

- |
- +--> 30 min rainfall
- +--> short-duration accumulation
- +--> 1h / 3h / 6h
- +--> 24h / 72h
- +--> intensity
- +--> duration

NASA SMAP

- |
- +--> antecedent wetness

SRTM DEM

- |
- +--> Flow direction
- +--> Flow accumulation
- +--> Catchment area
- +--> TWI
- +--> Drainage density
- +--> HAND
- +--> Valley geometry

Soil

- |
- +--> infiltration properties

Land cover

- |
- +--> imperviousness
- +--> runoff characteristics

Hydrography

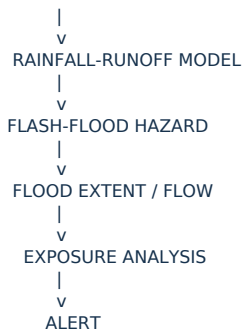
- |
- +--> streams
- +--> drainage network

Sentinel-1

- |
- +--> flood extent evidence

Historical flood inventory

- |
- +--> training labels



78. Compound Disaster Intelligence

The platform can combine:

Landslide hazard

+

Flash-flood hazard

+

Road exposure

+

Bridge exposure

+

Population exposure

+

Critical infrastructure

Example:

Heavy rainfall

|

+--> steep slope --> landslide hazard

```

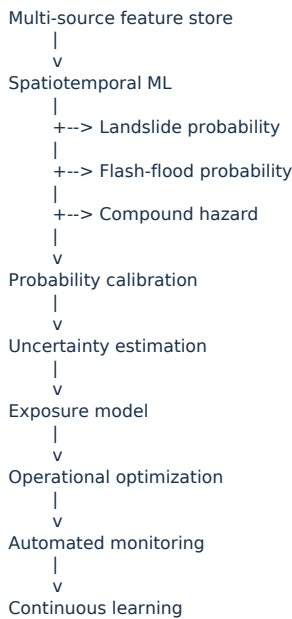
|
+--> saturated soil --> landslide hazard
|
+--> high flow accumulation --> flash-flood hazard
|
+--> low HAND --> flood exposure
|
+--> major road/bridge --> high operational impact

```

The result is a compound disaster response zone.

79. Advanced Future Architecture

After sufficient data accumulation:

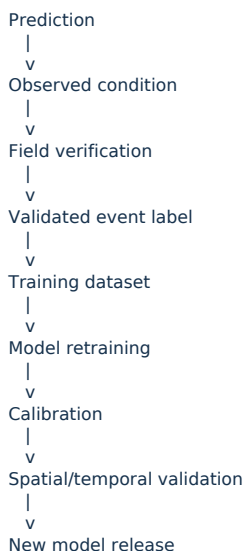


Potential advanced methods:

XGBoost / LightGBM
 Temporal Transformer
 Graph Neural Network
 ConvLSTM
 ensemble learning
 Bayesian uncertainty

80. Continuous Learning Loop

Every verified event becomes a new learning record:



This is how the prototype can progressively move toward a calibrated regional operational model.

81. Production-Grade Model Readiness

A genuinely calibrated landslide probability system requires:

Large validated event inventory
 + reliable background/non-event observations
 + temporal alignment
 + spatial coverage
 + multi-year observations
 + quality-controlled satellite data
 + spatial validation
 + temporal validation
 + independent test set
 + probability calibration
 + uncertainty estimation

The current pilot should therefore distinguish:

OPERATIONAL PROTOTYPE

from:

FULLY CALIBRATED REGIONAL PRODUCTION MODEL

The latter is achieved through data accumulation and repeated validation, not by assigning an arbitrary percentage to a prototype score.

82. Scientific Integrity Rules

The system must never:

- claim GPM is a 500 m native rainfall product;
- claim SMAP is a 500 m native soil-moisture product;
- claim a satellite observation is live when it is stale;
- convert missing data to zero;
- convert unknown cells into confirmed negatives;
- call an uncalibrated score a probability;
- claim exact deterministic landslide timing;
- claim SAR change is automatically a landslide;
- claim NDVI alone predicts landslides;
- treat roads/population as physical causes without engineering justification;
- claim flood depth without a validated hydraulic/hydrological basis;
- claim model accuracy without a valid independent evaluation;
- claim Cell Broadcast delivery without authorized infrastructure integration.

83. Current Pilot Implementation State

The existing prototype already contains the foundation for:

- 500 m analysis grid
- Kohima + Aizawl
- SRTM terrain
- slope
- elevation
- curvature
- TWI
- historical landslide evidence
- NASA GPM ingestion
- NASA SMAP ingestion
- dynamic risk engine
- MapLibre GIS
- cross-district area search
- cell-level details
- alert workflow
- officer verification
- CAP-ready alert structure
- data freshness gates
- FastAPI backend
- React frontend

Current dynamic source realities must remain visible:

GPM:

- real NASA observations
- near-real-time product
- freshness monitored

SMAP:

- real NASA observations
- daily product
- freshness monitored
- may be stale depending on latest acquisition

Sentinel-1:

- architecture ready
- actual data dependent on access/authentication

Sentinel-2:

- can provide vegetation/land-cover features once valid scenes are ingested

Flood:

- can use DEM hydrology + rainfall and later Sentinel-1 flood observations

Road/population/infrastructure:

- must be populated only from real datasets

84. Recommended Development Sequence

Phase 1 --- Stable foundation

- District boundaries
- 500 m grid
- SRTM
- terrain derivatives
- historical inventory

Phase 2 --- Dynamic rainfall

- GPM ingestion
- freshness
- rolling rainfall features
- rainfall duration
- antecedent rainfall

Phase 3 --- Soil moisture

- SMAP ingestion
- current/previous
- change
- freshness

Phase 4 --- Dynamic hazard

static susceptibility
+
rainfall
+
soil moisture

Phase 5 --- Optical/environmental

Sentinel-2
NDVI
NDMI
land cover
vegetation change

Phase 6 --- Radar

Sentinel-1
VV/VH
change
coherence/deformation

Phase 7 --- Hydrology

flow accumulation
drainage
HAND
rainfall-runoff
flood extent

Phase 8 --- Exposure

roads
bridges
population
settlements
buildings
critical infrastructure

Phase 9 --- ML

large validated inventory
PU -> supervised ML
spatial CV
temporal holdout
cross-district testing

Phase 10 --- Calibration

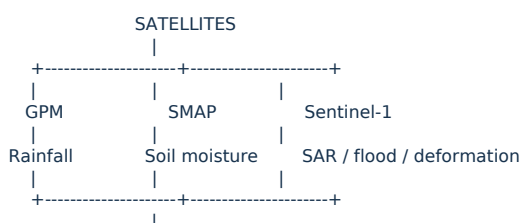
Platt / isotonic / beta calibration
Brier
ECE
reliability
independent test

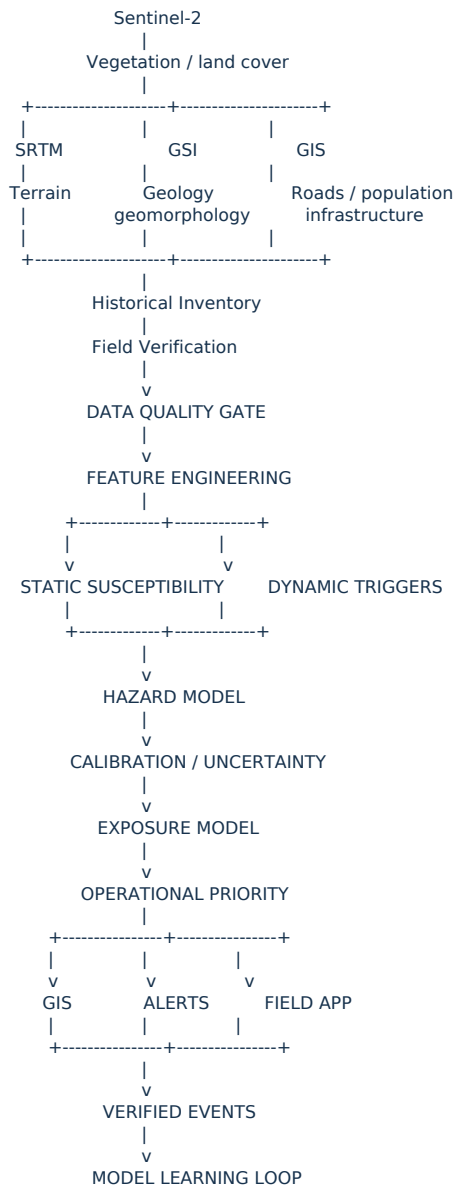
Phase 11 --- Operations

uncertainty
alerts
field verification
audit
monitoring
retraining

85. Final Technical Architecture

The complete target architecture is:





86. Reference Data Products and Documentation

NASA GPM

IMERG documentation:

<https://gpm.nasa.gov/data/imerg>

GPM data directory:

<https://gpm.nasa.gov/data/directory>

IMERG technical documentation:

<https://gpm.nasa.gov/resources/documents/imerg-v07-technical-documentation>

NASA SMAP / NSIDC

SMAP SPL3SMP_E Version 6:

https://nsidc.org/data/spl3smp_e/versions/6

NASA Earthdata Login:

<https://urs.earthdata.nasa.gov/>

NASA Earthdata Search:

<https://search.earthdata.nasa.gov/>

Sentinel-1

ESA Sentinel-1 mission:

https://www.esa.int/Applications/Observing_the_Earth/Copernicus/Sentinel-1/Introducing_Sentinel-1

ESA Sentinel-1 instrument:

https://www.esa.int/Applications/Observing_the_Earth/Copernicus/Sentinel-1/Instrument

Sentinel-2

ESA Sentinel-2 facts:

https://www.esa.int/Applications/Observing_the_Earth/Copernicus/Sentinel-2/Facts_and_figures

SRTM

USGS SRTM archive:

<https://www.usgs.gov/centers/eros/science/usgs-eros-archive-digital-elevation-shuttle-radar-topography-mission>

Geological Survey of India

GSI landslide hazard resources:

https://bhusanket.gsi.gov.in/LS_hazard.html

GSI Bhusanket:

<https://bhusanket.gsi.gov.in/>

OpenStreetMap

OSM licensing/use:

https://wiki.openstreetmap.org/wiki/License/Use_Cases

IMD / MAUSAM

Indian rainfall dataset reference:

<https://mausamjournal.imd.gov.in/index.php/MAUSAM/article/view/851>

87. One-Line Technical Summary

**SARVAS is a multi-source spatiotemporal geospatial system that
combines NASA GPM rainfall, NASA SMAP soil moisture, SRTM terrain,
Sentinel-1 SAR, Sentinel-2 vegetation, geological and hydrological
layers, historical landslides, field verification, and exposure data
to estimate dynamic landslide and flash-flood hazard, prioritize
affected assets, and support near-real-time disaster warnings.**

88. Key Engineering Principle

The final system should evolve as:

DATA
->
QUALITY
->
FEATURES
->
SUSCEPTIBILITY
->
DYNAMIC TRIGGERS
->
HAZARD
->
CALIBRATED PROBABILITY
->
UNCERTAINTY
->
EXPOSURE
->
PRIORITY
->
VERIFICATION
->
ALERT
->
OBSERVED EVENT
->
LEARNING

This is the complete A-to-Z technical path from the current SIH prototype to a scientifically defensible, calibrated, scalable regional disaster-intelligence platform.